Full Gradient Deep Reinforcement Learning for Average-Reward Criterion

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Abstract

We extend the provably convergent Full Gradient DQN algorithm for discounted reward Markov decision processes from [1] to average reward problems. We experimentally compare widely used RVI Q-learning with recently proposed Differential Q-learning in the neural function approximation setting with Full Gradient DQN and DQN. We also extend this to learn Whittle indices for Markovian restless multi-armed bandits. We observe a better convergence rate of the proposed Full Gradient variant across different tasks.

Preliminaries

Consider a controlled Markov chain $\{X_n, U_n\}, n \geq 0$, on a finite state space $\mathcal{S}$, controlled by a process $U_n, n \geq 0$, taking values in a finite action space $\mathcal{A}$, with transition probabilities $p(j|i,u) \in [0,1], i,j \in \mathcal{S}, u \in \mathcal{A}$, satisfying $\sum_j p(j|i,u) = 1 \; \forall \, i, u$. Its time evolution is described by

$$\Pr(X_{n+1} = j | X_m, U_m, m \leq n) = p(j|X_n, U_n).$$

Objective

Given a per-stage reward function $r : \mathcal{S} \times \mathcal{A} \mapsto \mathcal{R}$, the average reward MDP seeks to maximize the time-averaged reward

$$\beta^* = \liminf_{N \uparrow \infty} \mathbb{E}\left[\frac{1}{N}\sum_{n=0}^{N-1} r(X_n, U_n)\right].$$

For this, an optimality equation for Q-values is defined as

$$Q(i,u) = r(i,u) - \beta + \sum_j p(j|i,u) \max_v Q(j,v), \;\; i \in \mathcal{S}, u \in \mathcal{A},$$

where β is characterized uniquely as the optimal reward β^* and $Q(\cdot,\cdot)$ is uniquely characterized modulo an additive constant.

Bellman error

For a parameterized $Q(\cdot,\cdot;\theta)$, the optimality equation suggests minimizing following Bellman error

$$\mathcal{E}(\theta) := \mathbb{E}\Big[\Big(r(X_n,U_n) - f(Q;\theta) + \sum_j p(j|i,u) \max_v Q(j,v;\theta) - Q(X_n,U_n;\theta)\Big)^2\Big],$$

where $f(Q;\theta_n)$ is an offset, which for $f(Q) = Q(i_0,u_0), \max_u Q(i_0,u), \; \max_{i,u} Q(i,u), \; \frac{1}{s\ell}\sum_{i,u} Q(i,u)$, for some fixed $i_0 \in \mathcal{S}, u_0 \in \mathcal{A}$ can be shown to converge to β^* a.s.

The gradient of the Bellman error involves *product of conditional expectations*

$$\nabla \mathcal{E}(\theta) = 2\sum_i \pi(i)\Big[\Big(r(i,\pi(i)) - f(Q;\theta) + \sum_j p(j|i,\pi(i)) \max_v Q(j,v;\theta) - Q(i,\pi(i);\theta)\Big)$$
$$\times \Big(-\nabla f(Q;\theta) + \sum_j p(j|i,\pi(i)) \nabla Q(j,v';\theta) - \nabla Q(i,\pi(i);\theta)\Big)\Big]$$

where $v' \in \operatorname{Arg\,max}_v Q(j,v;\theta)$ and $\nabla Q(j,v';\theta)$ is a legitimate subgradient by Danskin's theorem.

Baird's Double-Sampling scheme

$$\theta_{n+1} = \theta_n - a(n)\Big(r(X_n,U_n) - f(Q;\theta_n) + \max_v Q(X_{n+1},v;\theta_n) - Q(X_n,U_n;\theta_n)\Big)$$
$$\times \Big(-\nabla f(Q;\theta_n) + \nabla Q(\tilde{X}_{n+1},\tilde{v}_{n+1};\theta_n) - \nabla Q(X_n,U_n;\theta_n)\Big)$$

where $\tilde{v}_{n+1} \in \operatorname{Arg\,max}_v Q(\tilde{X}_{n+1},v;\theta)$ and $\tilde{X}_{n+1} \sim p(\cdot|X_n,U_n)$ drawn conditionally independent of $X_{n+1} \sim p(\cdot|X_n,U_n)$

- ▷ For stochastic problems, this scheme considers sampling two independently drawn samples from the simulator for a single update iteration. This may not be possible in reality unless we have full access to the simulator.

Algorithms

Full Gradient Deep Q-Network (FGDQN)

Our algorithm does not require double-sampling and resembles the stochastic gradient descent for the error $\mathcal{E}(\theta)$. It involves the following update step

$$\theta_{n+1} = \theta_n + a(n)\Big(r(X_n,U_n) - f(Q;\theta_n) + \max_v Q(X_{n+1},v;\theta_n) - Q(X_n,U_n;\theta_n)\Big)$$
$$\times \Big(\nabla f(Q;\theta_n) - \nabla Q(X_{n+1},v_{n+1};\theta_n) + \nabla Q(X_n,U_n;\theta_n)\Big)$$

where $v_{n+1} \in \operatorname{Arg\,max}_v Q(X_{n+1},v;\theta)$ and the overline denotes empirical average over the triplets $(X_m,U_m,X_{m+1}), m < n$ for which $X_m = X_n$ and $U_m = U_n$.

FGDQN algorithm handles the first conditional expectation by taking an empirical average using the experience replay and the second conditional expectation is handled by the stochastic approximation.

Algorithm:
FGDQN update step for triplet (i,u,j)

1. Sample the triplets with same state i and control u from the experience replay
2. Compute the empirical average
3. Compute the gradient for the triplet (i,u,j)
4. Update the parameters θ

Deep Q-Network (DQN)

We consider a natural extension of the DQN algorithm [2] for average reward, viz.,

$$\theta_{n+1} = \theta_n + \frac{a(n)}{B} \times \sum_{b=1}^{B}\Big((Z_{n(b)} - Q(X_{n(b)},U_{n(b)};\theta_n))\nabla_\theta Q(X_{n(b)},U_{n(b)};\theta_n)\Big),$$

where $(X_{n(b)},U_{n(b)},X_{n(b)+1}), 1 \leq b \leq B$ are randomly selected triplets from the experience replay, and

$$Z_{n(b)} = r(X_{n(b)},U_{n(b)}) + \max_v Q(X_{n(b)+1},v;\tilde{\theta}_n) - f(Q;\tilde{\theta}_n)$$

is the 'target' being chased by $Q(X_{n(b)},U_{n(b)})$. $\tilde{\theta}_n$ is updated on a slower time scale by setting it equal to θ_n periodically and left unaltered in between. As pointed out in [1], there are several theoretical issues about DQN, but it remains a popular scheme because of good empirical behavior.

Restless Bandits

The problem of Markovian restless bandits considers N arms taking two actions, 'active' and 'passive.' The objective is to maximize the average reward subject to the constraint that every round $M < N$ arms have to be active. [3] proposed an index policy to solve this problem by relaxing the per-stage constraint to the average constraint

$$\liminf_{n \uparrow \infty} \mathbb{E}\left[\frac{1}{n}\sum_{m=0}^{n-1}\sum_{i=1}^{N} \nu_m^i\right] = M.$$

Whittle solves this problem using the Lagrange multiplier λ, which is related to the subsidy for passive action. For each arm, a Whittle index λ^* is defined as the infimum subsidy to pay so that it is equally desirable to give the arm active or passive action. The Whittle index policy computes and sorts the Whittle index for each arm and activates top M arms.

Based on this intuition [4] proposed the Whittle index as the threshold for which the following holds

$$Q(\hat{k},1) = Q(\hat{k},0)$$

for each arm $\hat{k}$ where the dependence of $Q(\hat{k},\cdot)$ on λ is implicit.

We consider the problem of large MDPs, and characterize λ by a parameterized family $(\hat{k},\sigma) \to \lambda(\hat{k};\sigma)$ and Q-values by a parameterized family $(x,u,\lambda(\hat{k};\sigma),\theta) \to Q(x,u,\lambda(\hat{k};\sigma);\theta)$ where $x \in \mathcal{S}$ and $u \in \{0,1\}$. This is done to render explicit the dependence of Q on $\lambda(\hat{k})$ for each $\hat{k}$ separately.

FGDQN for Whittle Index Learning

The proposed algorithm is a two-time scale where

- ▷ on a faster time scale: learn $Q(j,u,\lambda(\hat{k};\sigma);\theta_n), j \in \mathcal{S}, u \in \{0,1\}$ values with modified reward as

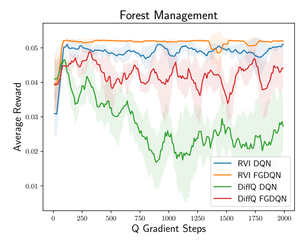
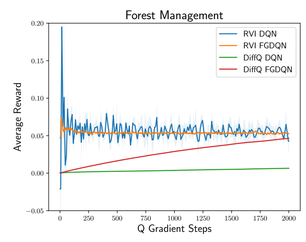
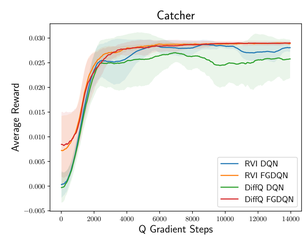
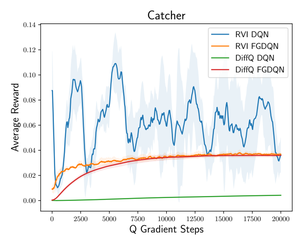
$$\tilde{r}(X_n,U_n;\hat{k}) = (1-U_n)(r_b(X_n) + \lambda(\hat{k};\sigma)) + U_n r_a(X_n)$$

  and update θ *for each* $\hat{k} \in \mathcal{S}$ *separately* using FGDQN or DQN.
- ▷ on a slower time scale: minimize the following error and update σ using stochastic gradient descent

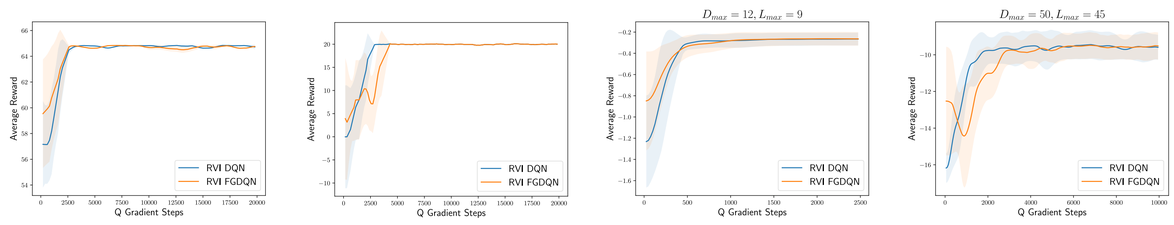
$$\bar{\mathcal{E}}(\sigma) = \mathbb{E}\Big[\Big(Q(\hat{k},1,\lambda(\hat{k};\sigma);\theta) - Q(\hat{k},0,\lambda(\hat{k};\sigma);\theta)\Big)^2\Big].$$

Experimental Results

Here we consider infinite-horizon problems with average-reward objective. Following we plot average reward and reward proxy ($f(Q)$ for RVI and $\bar{R}$ for DiffQ learning) during evaluation i.e. after every K Q-gradient steps for suitably chosen K.

Problem description	Evaluation average reward	Average reward proxy
Forest Management The task is to maintain an old forest for wildlife and make money by selling the cut wood with the possibility of fire burning the forest. The state is the age of the forest with two actions of 'cut' and 'no cut' and the reward is the current age for cut and 0 for no cut.		
Catcher The task involves catching a fruit with a paddle in a grid where the fruit falls from top to bottom in about 40 time steps. The state is the paddle's position, velocity, and position of the fruit. Action is to move the paddle left or right with a reward of 1/-1 for a successful/unsuccessful catch.		

For restless bandits, we consider problems of circulant and restart Markov chains motivated by applications in congestion control, web crawling, and machine maintenance. We also consider the problem of deadline scheduling which involves scheduling Electrical Vehicles (EVs) to charging stations based on the amount of charging needed and the deadline. Following are the experimental results



Circulant Dynamics | Restart Problem | Deadine Scheduling

Evaluation average reward

References

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